

Project Type: SystemVerilog RTL Design and Verification Project

Author: Narasimhamurthy B

32-bit Comparator with Interrupt Generation

1. Overview

The Comparator module is a 32-bit equality comparator designed for interrupt generation applications in digital systems.

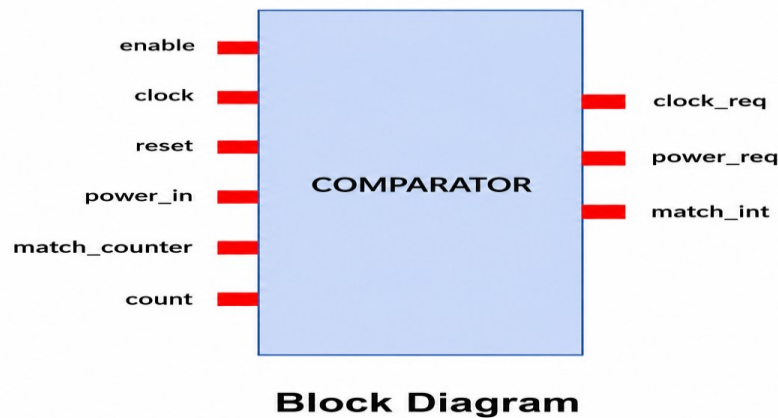
The module compares an incoming 32-bit count value with a programmable 32-bit match_counter value. When both values are equal, the module asserts the match_int signal.

The design also supports:

- Enable-controlled operation
- Power-aware functionality
- Asynchronous reset support
- Synchronous interrupt generation

This module is suitable for timer systems, event detection, programmable comparison logic, and interrupt-driven digital applications.

2. Block Diagram



3. Key Features

3.1 32-bit Equality Comparison

- Compares two 32-bit input values.
 - Equality comparison is performed between:
 - count
 - match_counter
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3.2 Interrupt Generation

- match_int is asserted when:

count == match_counter

- Interrupt is automatically de-asserted when values differ.
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3.3 Enable-Controlled Operation

- Comparator operates only when enable = 1.
 - If enable = 0, the interrupt output is cleared.
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3.4 Power-Aware Operation

- Comparison logic is active only when power_in = 1.
 - If power_in = 0, interrupt output is cleared.
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3.5 Asynchronous Reset Support

- Active-high asynchronous reset.
 - Reset clears match_int immediately.
 - Supports safe startup and recovery conditions.
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4. Signal Description

Signal Name	Direction	Width	Description
clk	Input	1	System clock
reset	Input	1	Asynchronous reset
enable	Input	1	Enables comparator operation
power_in	Input	1	Power enable signal
match_counter	Input	32	Programmable reference value
count	Input	32	Input value for comparison
match_int	Output	1	Match interrupt output

5. Functional Behavior

Comparator Operation

Case 1: Reset Active

If:

reset = 1

Then:

match_int = 0

Case 2: Enable Disabled

If:

enable = 0

Then:

match_int = 0

Case 3: Power Disabled

If:

```
power_in = 0
```

Then:

```
match_int = 0
```

Case 4: Normal Comparison Mode

If:

```
enable = 1  
power_in = 1
```

Then:

```
if(count == match_counter)  
    match_int = 1  
else  
    match_int = 0
```

6. RTL Design Methodology

The comparator module was designed using Verilog HDL.

The design uses:

- Sequential always block
 - Positive edge-triggered clocking
 - Asynchronous reset handling
 - Conditional interrupt generation logic
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7. Verification Methodology

The module was verified using a constrained-random SystemVerilog verification environment.

Verification Components

- Generator
 - Driver
 - Monitor
 - Scoreboard
 - Functional Coverage
 - Interface with clocking blocks
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Verification Features

- Constrained-random stimulus generation
 - Mailbox-based communication
 - Scoreboard-based result checking
 - Functional coverage collection
 - Pass/Fail reporting
 - Waveform analysis using EPWave and QuestaSim
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8. Functional Coverage

Functional coverage was implemented to verify:

- Enable conditions
 - Power conditions
 - Equality comparison scenarios
 - Mismatch scenarios
 - Interrupt assertion behavior
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9. Applications

- Timer-based event triggering
 - Interrupt generation systems
 - Counter comparison logic
 - Embedded digital controllers
 - FPGA and ASIC subsystem designs
 - Programmable monitoring systems
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10. Conclusion

The 32-bit Comparator module successfully performs equality-based comparison and interrupt generation with support for enable control, power-aware operation, and asynchronous reset handling.

The module was verified using a modular SystemVerilog constrained-random verification environment with scoreboard checking and functional coverage collection.

The design is suitable for FPGA, ASIC, and digital subsystem integration applications.